Recall

Work from memory, then check with your tutor.

R1. Red : blue is 3 : 5. How many blue tiles go with 6 red tiles?

Answer: _____

R2. A snail moves 3 centimeters each minute. Find the distance after 0, 4, and 7 minutes.

Answer: _____

R3. Find $\frac{3}{5} \div \frac{1}{5}$.

Answer: _____

R4. Complete the ratio-table pair: (3, 4), (12, __).

Answer: _____

S. Return here near the end of the lesson

To keep a quotient unchanged, I multiply both ___ and ___ by the same number.

Long division with decimals

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Use long division to find 672 ÷ 24.

Answer: _____

P2. Find 27 ÷ 8. Continue into decimal places.

Answer: _____

P3. Find 8.4 ÷ 7.

Answer: _____

Long division with decimals

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Find 6.3 ÷ 0.09 by writing an equivalent division with a whole-number divisor.

Answer: _____

P5. Find 7 ÷ 20.

Answer: _____

P6. A 3.6-meter ribbon is cut into 0.4-meter pieces. How many pieces?

Answer: _____

Long division with decimals

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. Find 1,176 ÷ 28 and check by multiplication.

Answer: _____

F. Complete: 5.4 ÷ 0.03 = __ ÷ 3 = __.

Answer: _____

D. Lina changes 6.3 ÷ 0.09 into 6.3 ÷ 9 and gets 0.7. Which change is missing?

Answer: _____

Mixed review

M1. A mix has 5 cups oats and 3 cups seeds. What fraction of the whole is seeds?

Answer: _____

M2. Find 12% of $15.00.

Answer: _____

M3. Find $2\frac{1}{2} \div \frac{3}{4}$.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. Find 8.4 ÷ 0.07. Write an equivalent division and check by multiplication.

Answer: _____

Decimals and long division

The big idea

Decimal digits name parts of a whole. Addition and subtraction combine equal place-value units. Division preserves its quotient when both quantities scale equally.

Words to know

Place value: A digit’s value depends on its position. Example: The 4 in 0.04 means four hundredths.

Estimate: A nearby value that helps check an answer. Example: 6.2×3 is close to $6 \times 3 = 18$.

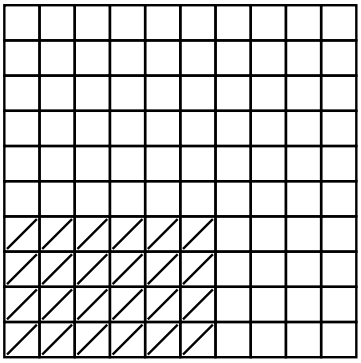
Thousandth: One of 1,000 equal parts of a whole. Example: 0.007 means seven thousandths.

Remainder: The amount left after equal groups are made. Example: $17 \div 5$ gives 3 groups with 2 left.

Equivalent division: A new dividend and divisor with the same quotient. Example: $6.4 \div 0.8 = 64 \div 8$.

Decimal quotient: A division result with fractional places written in decimals. Example: $7 \div 2 = 3.5$.

How it works



$0.6 \times 0.4 = 24$ hundredths. Addition combines like-sized units. Multiplication combines both the number of parts and the size of each part.

dividend	divisor
4.8	0.06
480	6

Multiply both quantities by 100. The quotient stays 80. Multiplying the dividend and divisor by the same power of ten preserves the quotient.

Worked examples

1. Find 3.8 + 0.475.

Write 3.800 + 0.475.

Add thousandths to thousandths and tenths to tenths.

3.800 + 0.475 = 4.275. The sum is a little above 4.

Answer: 4.275

2. Find 4.8 ÷ 0.06.

Rename both quantities in hundredths.

4.8 is 480 hundredths; 0.06 is 6 hundredths.

480 ÷ 6 = 80. Check 80 × 0.06 = 4.8.

Answer: 80

3. Long division: 936 ÷ 24

24 fits into 93 three times. Write 3 above the tens place. Subtract 72 from 93 to leave 21. Bring down the 6 to make 216. 24 fits into 216 nine times. Write 9 above the ones place. Subtract 216 to leave 0. The quotient is 39. Check: 39 × 24 = 936.

Common mistakes

Aligns the rightmost digits. Why: Treats every digit as the same unit. Fix: Write zeros to make place-value columns visible.

Makes a product larger than both small factors. Why: Keeps whole-number multiplication intuition. Fix: Take a part of one shaded part of a unit square.

Moves only one decimal point. Why: Uses a memorized instruction without equivalence. Fix: Rename both quantities in the same unit.

Try it

1. Three notebooks cost $2.45 each. What is the total cost?

2. A ball costs $4.65 and a card costs $2.78. Find the change from $10.

3. Find $9.6 \div 8$.

4. Find $9 \div 40$.

5. Find $1{,}344 \div 32$. Verify by multiplication.

Answers

1. $7.35 2. $2.57 3. 1.2 4. 0.225 5. 42

Where this connects

This builds on whole-number operations and fraction groups. Accurate decimal arithmetic supports expressions and equations next week.